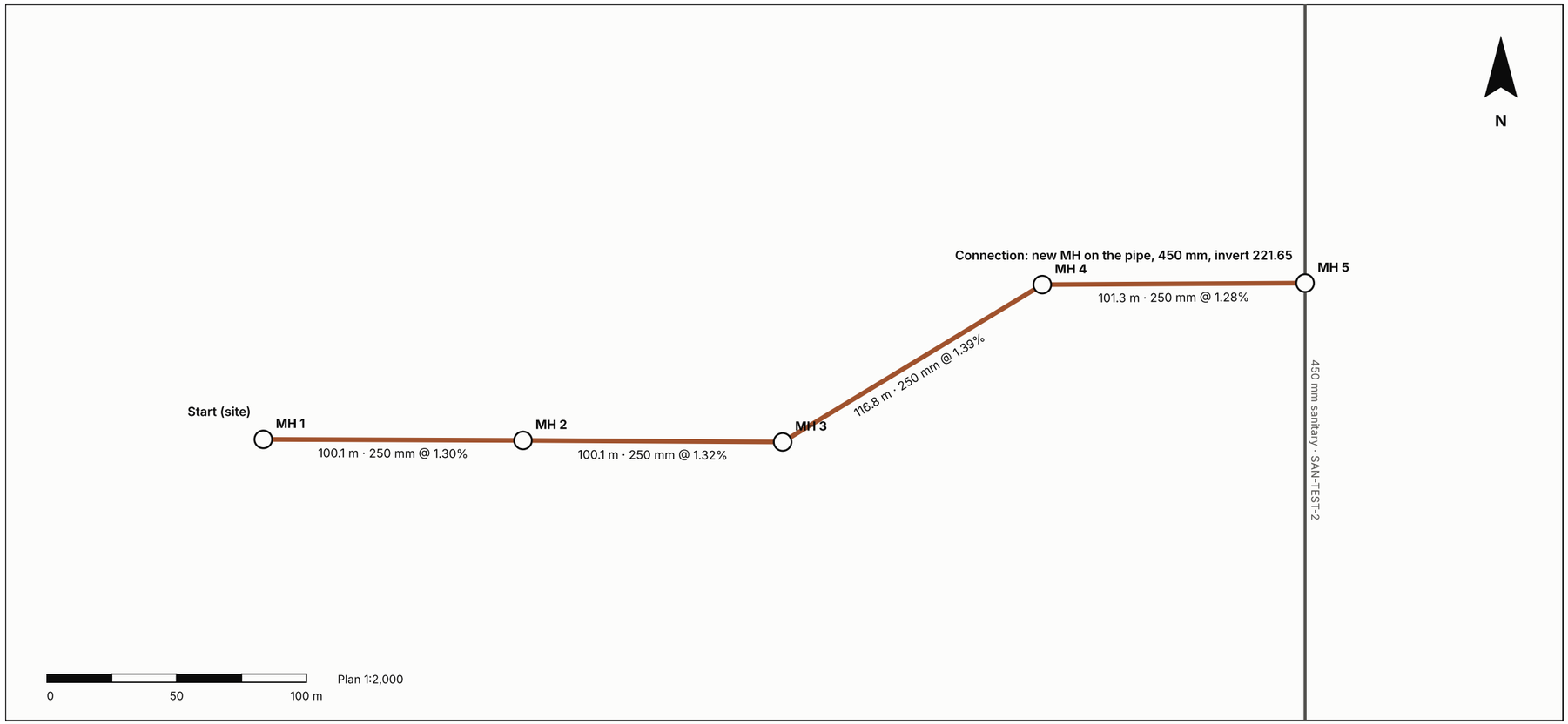
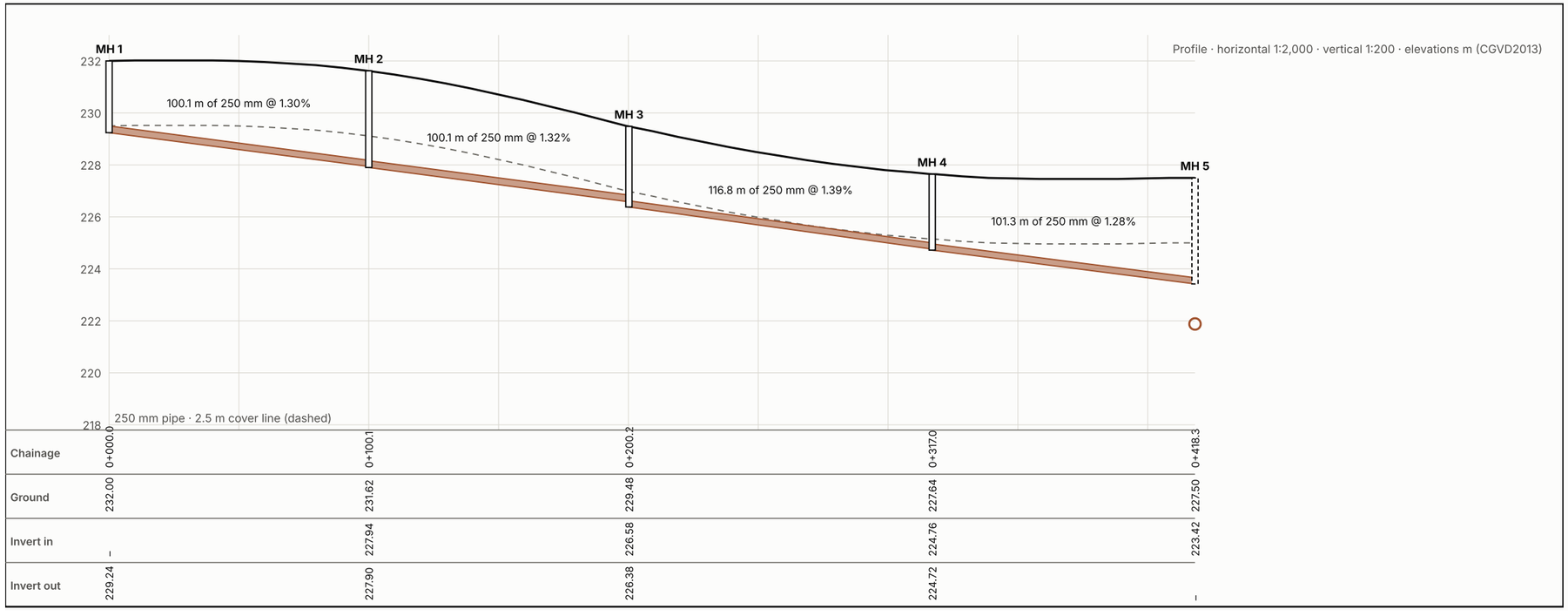




DESIGN

A **250 mm** sewer at 1.28–1.39%, 5 maintenance holes, 2.8 m deep at the start and 4.1 m at the deepest, for 47.1 L/s (36.7 L/s peak dry + 10.4 L/s I&I): the least excavation the rules allow.

LEGEND

- The new sewer (250 mm)
- The existing sanitary sewer it connects to
- Other Region sanitary sewers
- Maintenance hole (numbered from the start)
- Ground (profile)
- 2.5 m cover line (profile)
- Drop structure (dashed maintenance hole)

RULES CHECKED

- ✓ Pipe size (200 mm minimum, no larger than the existing sewer): 250 mm into 450 mm
- ✓ Velocity at the design flow (0.75 m/s or more): 1.48 m/s at the slowest
- ✓ Velocity flowing full (3.0 m/s or less): 1.43 m/s at the fastest
- ✓ Flow ratio, design flow to full flow (0.7 or less): 0.70
- ✓ The last leg, at the upstream end (1.00% or more): 1.30%
- ✓ Cover over the pipe (2.5 m or more): 2.50 m at the least
- ✓ Trench (8.0 m or less): 4.23 m at the deepest
- ✓ Maintenance holes (200 m apart or less): 117 m at the longest
- ✓ Drops through the maintenance holes (0.03–0.20 m, or a drop structure from 0.9 m): 1 drop structure
- ✓ Connection (the new obvert at or above the existing one): 1.57 m drop (drop structure)

4 reaches with supercritical flow (Froude 1 or more): the standard asks for subcritical flow where possible (s. 2.9.1).

PRELIMINARY · NOT FOR CONSTRUCTION

Preliminary sanitary sewer Plan and profile

Location	From 43.66999, -79.84262
Connection	450 mm sanitary sewer SAN-TEST-2, invert 221.65 m (new maintenance hole on the pipe)
Design flow	47.1 L/s (36.7 L/s peak dry + 10.4 L/s I&I)
Sewer	250 mm at 1.28–1.39%, 5 maintenance holes, 418 m; 2.8 m deep at the start, 4.1 m at the deepest; 1 drop structure
Scales	Plan 1:2,000 · profile 1:2,000 H, 1:200 V, on 11×17 at 100%
Ground	Ontario lidar terrain model (Test lidar, 0.5 m) (not a survey)
Sewers	Region of Peel sanitary mains (open data), read October 17, 2026
Rules	Region of Peel Linear Wastewater Standards (published Nov 2022, R1.0 Mar 2023), s. 3.1–3.4 and 6.1

General screening results only. Peel's W&WW Infrastructure Planning team must validate or confirm all results.

October 17, 2026 **Sheet 1 of 2**

Unofficial prototype · public data only · not a Region of Peel product · Labs · vdev

MAINTENANCE HOLES

MH	Chainage	Ground (m)	Invert in (m)	Invert out (m)	Depth to invert (m)	Drop (m)	
MH 1 (start)	0+000.0	232.00	–	229.24	2.76	–	
MH 2	0+100.1	231.62	227.94	227.90	3.72	0.04	
MH 3	0+200.2	229.48	226.58	226.38	3.10	0.20	
MH 4	0+317.0	227.64	224.76	224.72	2.92	0.04	
MH 5 (connection)	0+418.3	227.50	223.42	–	4.08	1.57	drop structure

REACHES

Reach	Diameter (mm)	Length (m)	Slope (%)	Invert up (m)	Invert down (m)	Full flow (L/s)	Flow ratio	V design (m/s)	V full (m/s)	Depth ratio	Froude
MH 1 → MH 2	250	100.1	1.30	229.24	227.94	67.8	0.70	1.49	1.38	0.61	1.32
MH 2 → MH 3	250	100.1	1.32	227.90	226.58	68.3	0.69	1.50	1.39	0.61	1.34
MH 3 → MH 4	250	116.8	1.39	226.38	224.76	70.0	0.67	1.53	1.43	0.60	1.38
MH 4 → MH 5	250	101.3	1.28	224.72	223.42	67.4	0.70	1.48	1.37	0.62	1.31

NOTES

- A preliminary design for screening: the route as drawn, the least excavation the rules allow, one pipe size along the run. Not a detailed design and not for construction.
- Ground from Ontario lidar terrain model (Test lidar, 0.5 m), not a survey; cover is measured from the ground along the line, not the road centreline (s. 3.4.1.1). The existing sewer's invert and size are as the Region publishes them.
- Design: Manning's n 0.013; 0.75 m/s or more at the design flow; 3.0 m/s or less flowing full; design flow 0.7 of full flow or less; the last leg 1.00% or more; 2.5 m of cover over the pipe; trench 8.0 m or less (0.15 m bedding); maintenance holes 200 m apart or less (190 m from 600 mm); drops 0.03–0.20 m, or a drop structure from 0.9 m; the new obvert at or above the existing one.
- Not considered: utilities and crossings, soils, groundwater, easements, construction method, lands external to the site (s. 3.4.1.1).
- Drawn without a base map. Print at 100% (no "fit to page") on 11×17 landscape to keep the scales; the scale bar holds at any size.

General screening results only. Peel's W&WW Infrastructure Planning team must validate or confirm all results.